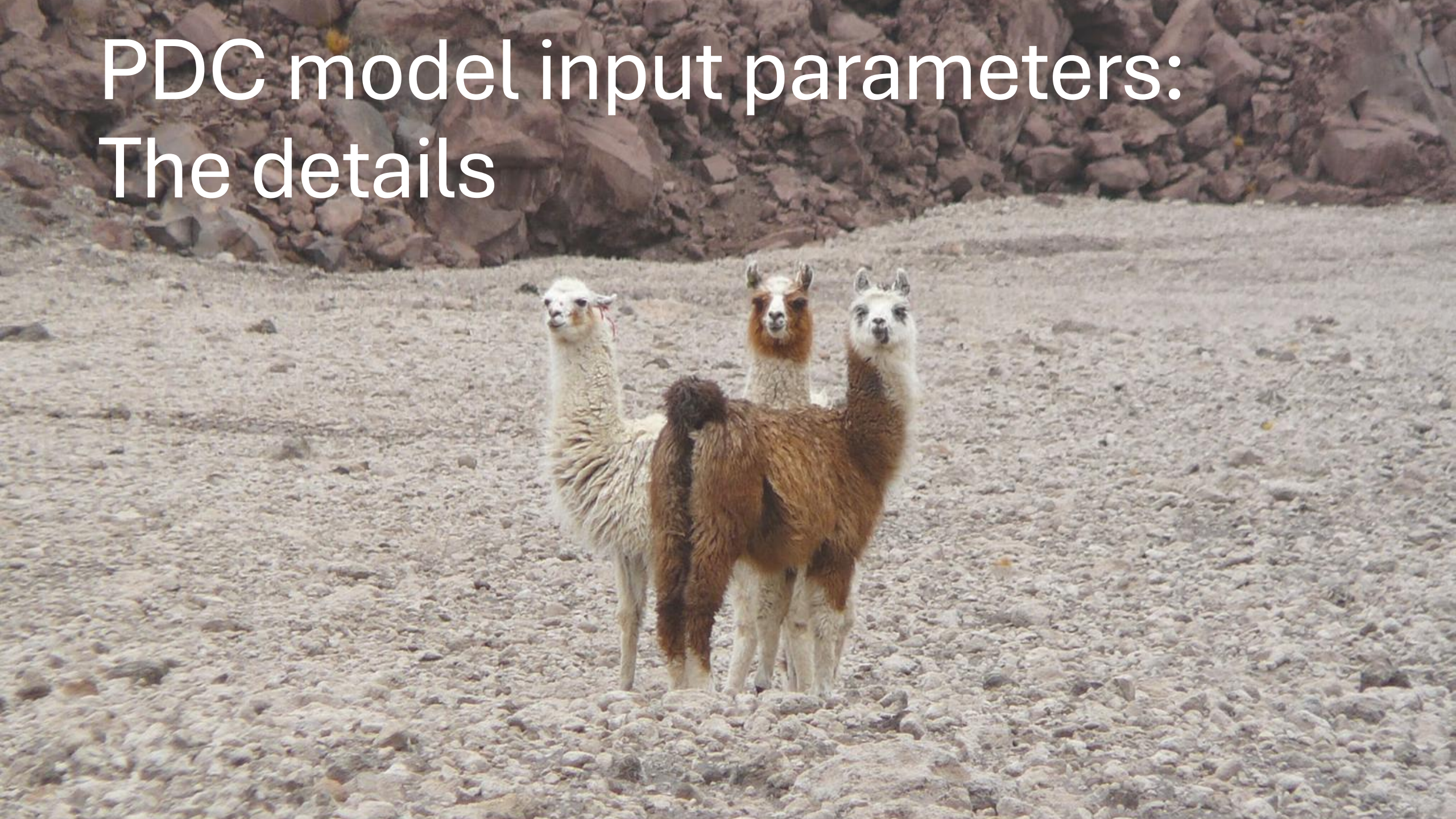
PDC model input parameters: The details



What do we need to know for modeling PDCs?

- Volume:
  - Dome-collapse: volume of dome, is dome growing, past deposits, global data
  - Column-collapse: volume of explosion scenarios, volume partitioning data, past deposits, global data
- Rheology/Mobility: several models use ΔH/L or basal friction angles that are a measure of PDC mobility and are related to flow rheology
  - These are closely related to volume though! So once you decide on volumes, you can just calculate with ΔH/L or basal friction angles to use
  - Past deposits, regressions from global data
- Source shape parameters and orientation
- Velocity
- Direction:
  - radial (column-collapse) or not (dome-collapse), past deposits

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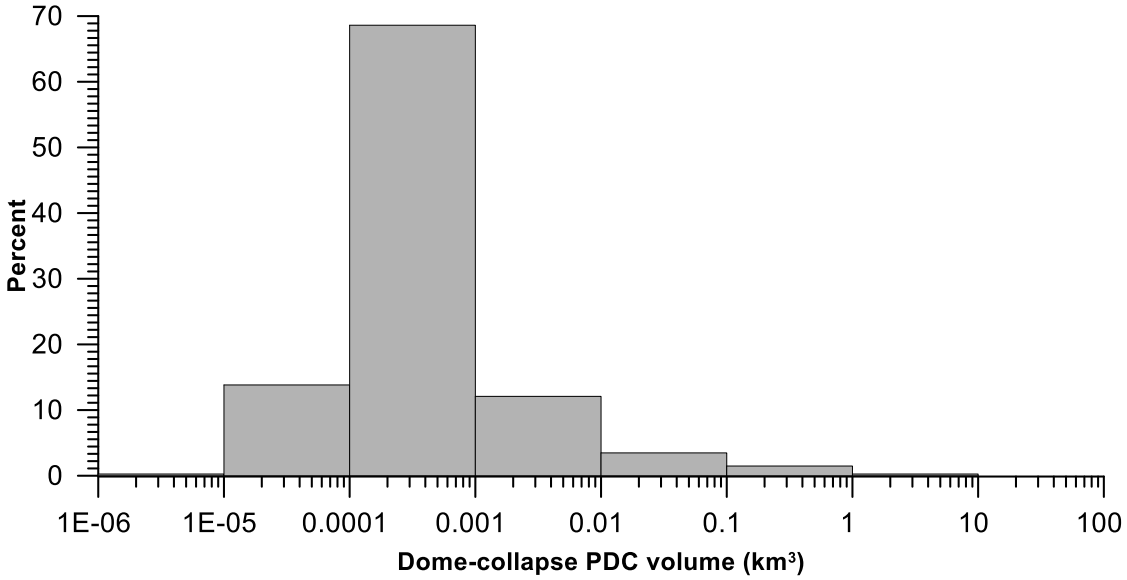
Energy cone only needs ΔH/L calculated from volume

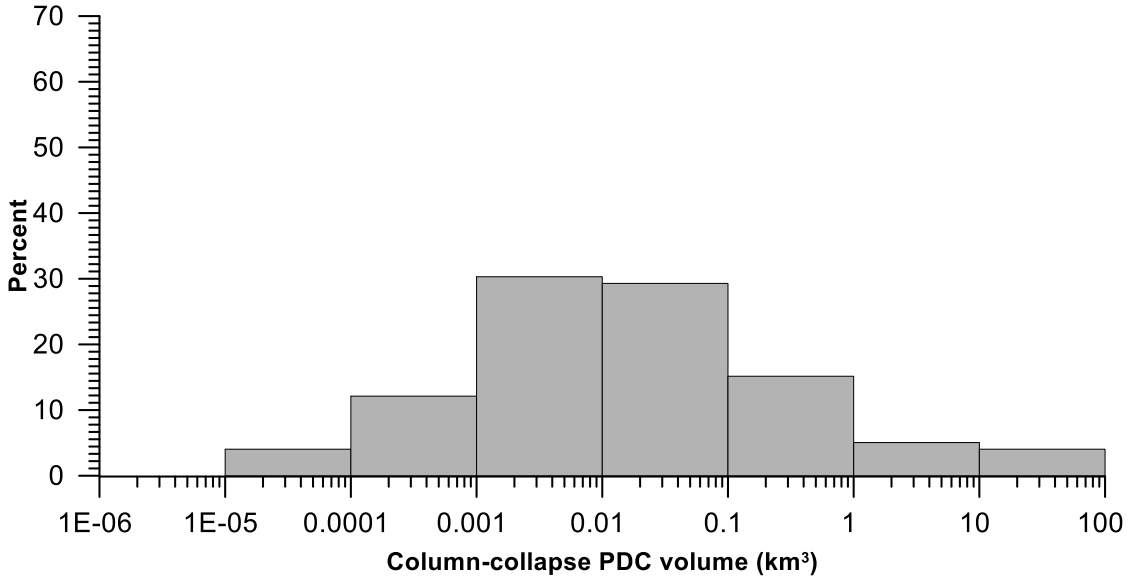
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Volume



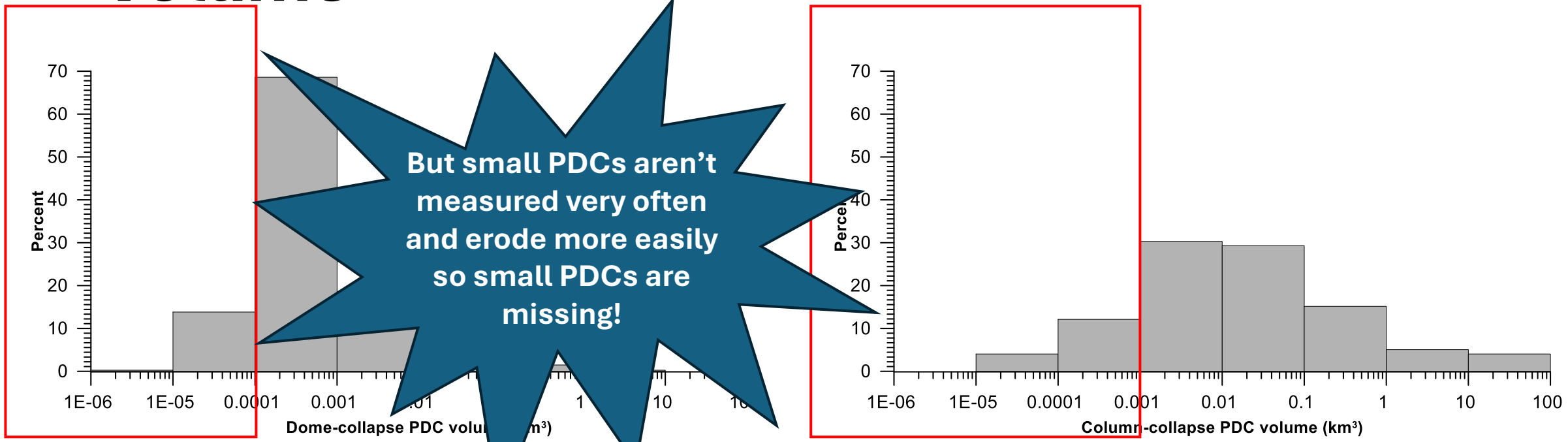


- Dome-collapse PDC volume is generally limited by the size of the dome
- Most dome-collapse PDCs are 0.0001-0.001 km^3 (0.1-1 M m^3)

- The column-collapse PDC volume distribution is more spread out
- Most column-collapse PDCs are 0.001-0.01 km^3 (1-10 M m^3)

Data from FlowDat (Ogburn 2025)

Volume



But small PDCs aren't measured very often and erode more easily so small PDCs are missing!

- Dome-collapse PDC volume is generally limited by the size of the dome
- Most dome-collapse PDCs are 0.0001-0.001 km^3 (0.1-1 M m^3)

- The column-collapse PDC volume distribution is more spread out
- Most column-collapse PDCs are 0.001-0.01 km^3 (1-10 M m^3)

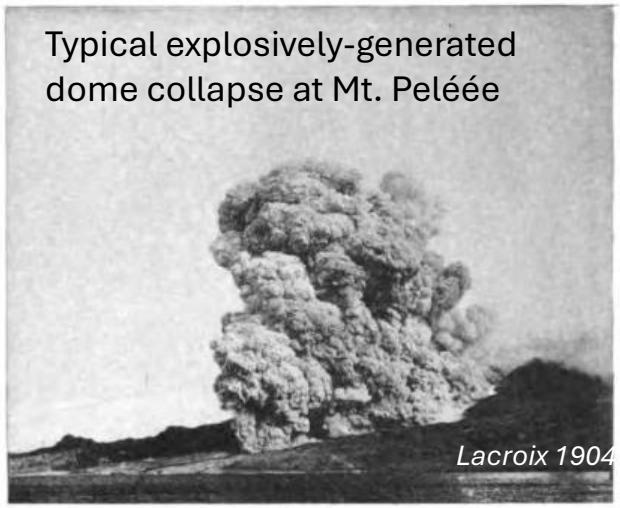
Data from FlowDat (Ogburn 2025)

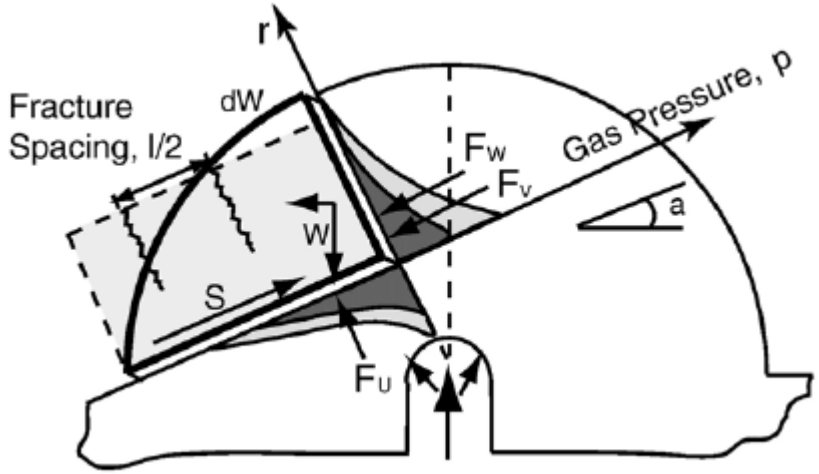
What contributes to frequency-volume distributions of dome-collapse PDCs?

- Size of the dome
- Dome extrusion rate
- Explosive activity
- Over-steepening, undercutting, other factors causing instability
- Extreme rainfall

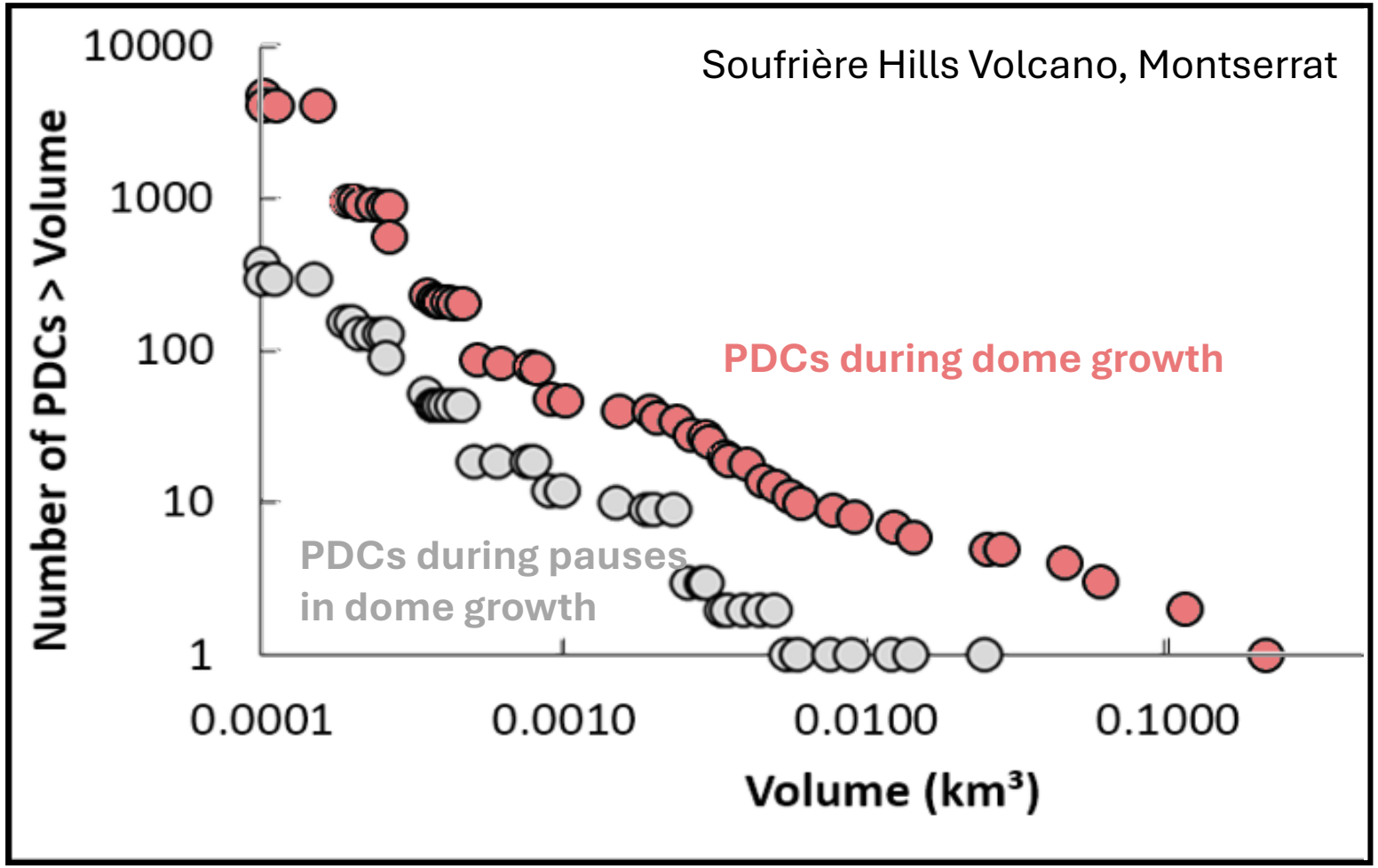






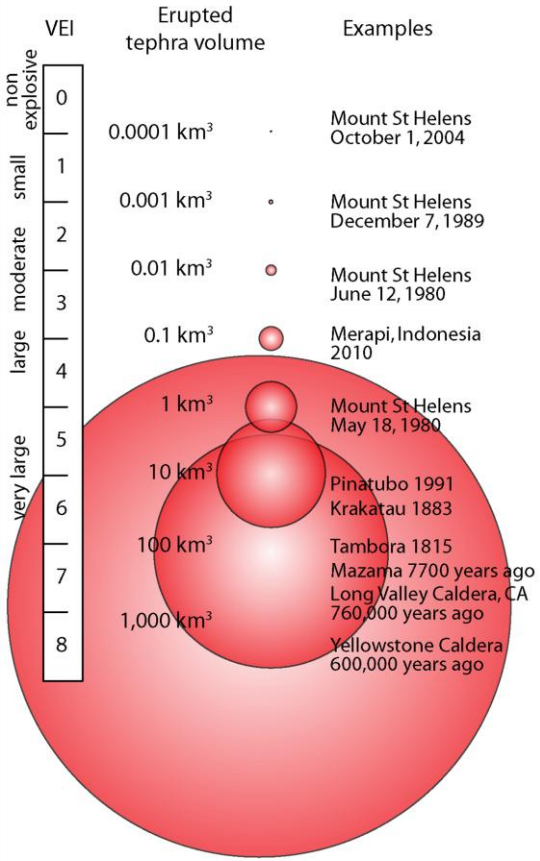


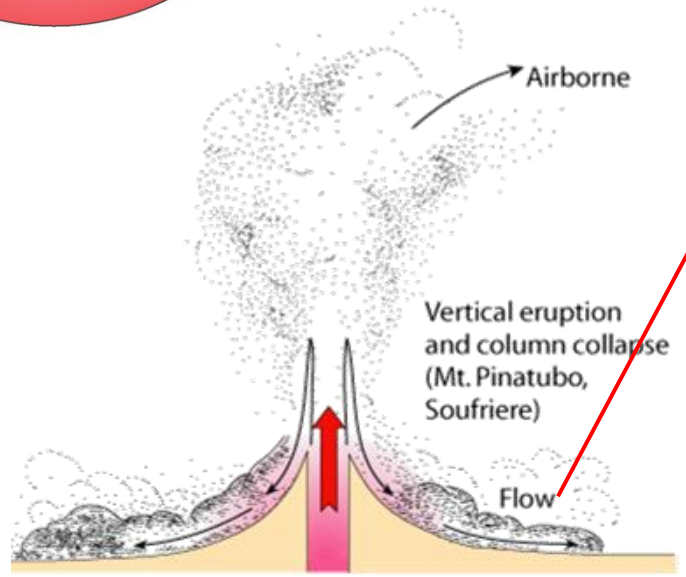
Model for rain-triggered dome collapse, Taron et al. 2007

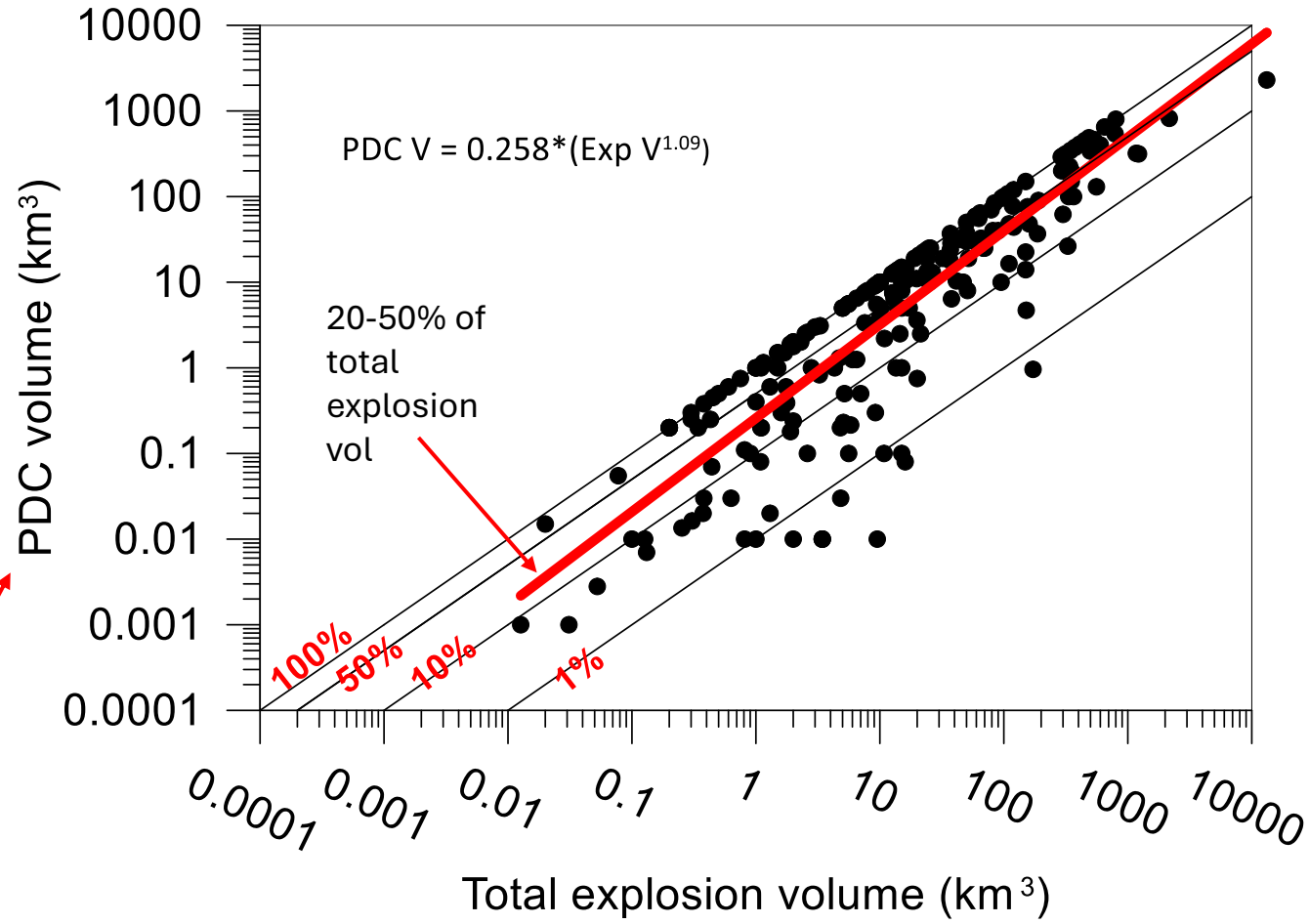


Frequency-magnitude plots from **FlowDat (Ogburn, 2025),** sorted by periods of dome growth from **DomeHaz** show that PDCs are both more frequent and larger in volume during **dome growth** at Montserrat

How big are column-collapse PDCs for a given VEI?
How much erupted volume collapses to form PDCs?



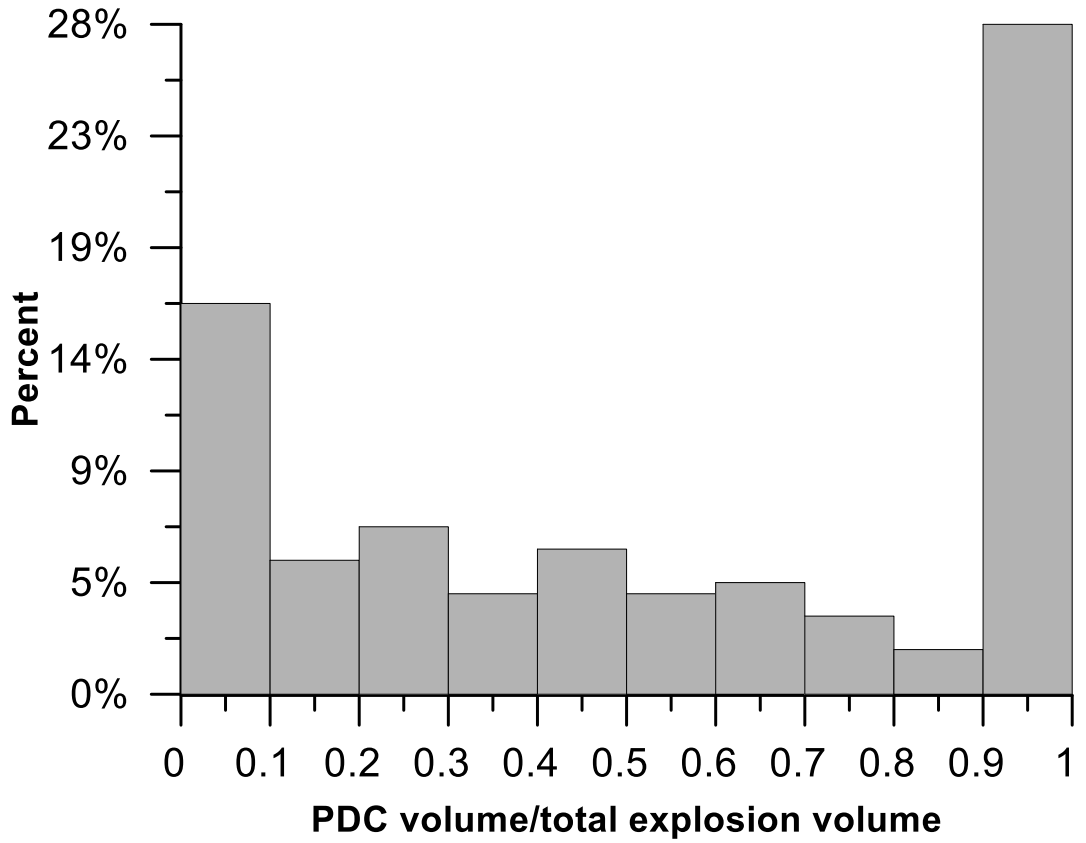




Data from LaMEVE (Crossweller et al. 2012) and FlowDat (Ogburn 2025)

If the most or all of the column collapsed (worst case scenario)

VEI	Max volume (km3)	Max volume (M m3)
1	0.001	1
2	0.01	10
3	0.1	100
4	1	1000
5	10	10,000
6	100	100,000
7	1000	1,000,00

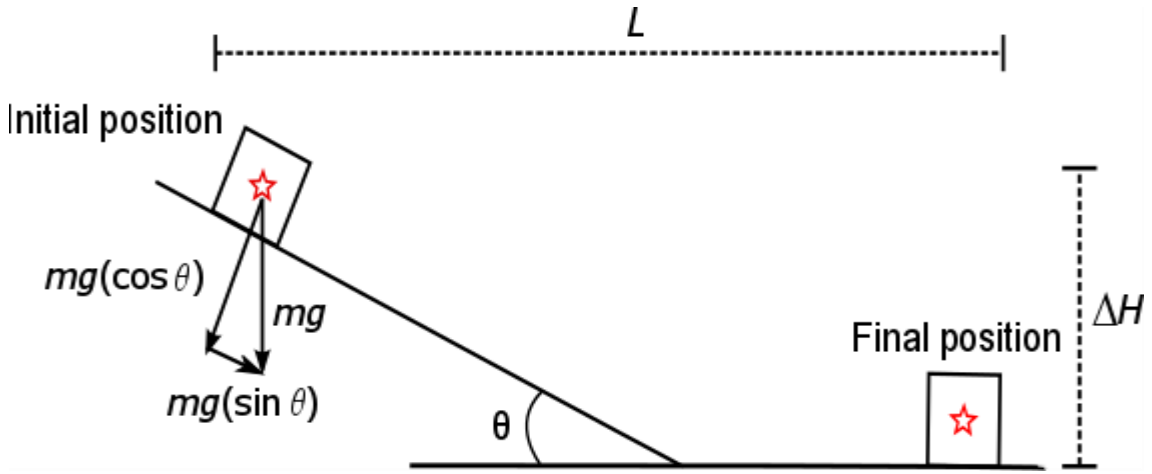


Flow Rheology

- Mohr-Coulomb Friction (sliding block model of friction, dry friction)
- Voellmy-Salm (adds turbulent friction in addition to Mohr-Coulomb friction)
- Pouliquen-Forterre (visco-plastic rheology)
- Pitman-Le (for water-saturated flows like lahars, uses two-phase flow)

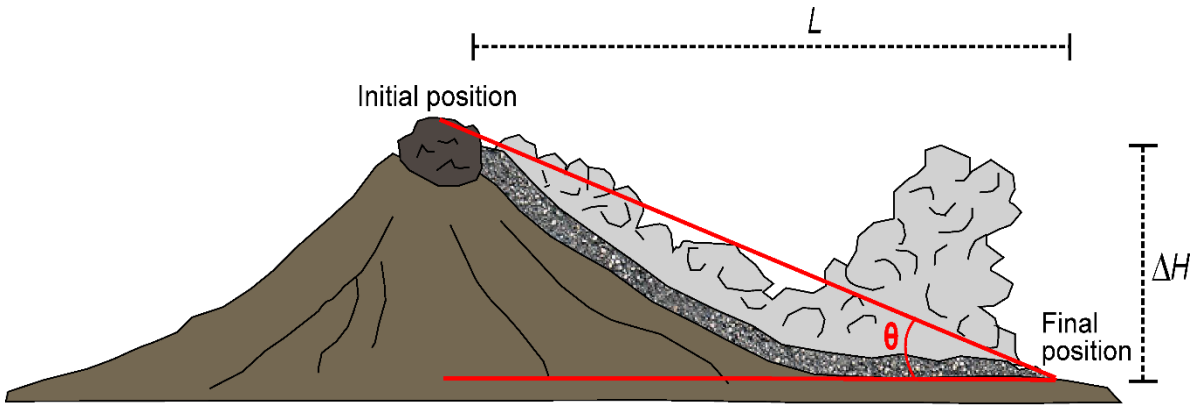
Flow Rheology

- Mohr-Coulomb Friction
- Block begins to slide when gravitational force down a plane equals the frictional force
- when mgsinθ = μmgcosθ
- μ = Tanθ =coeff. of friction
- coeff. of friction= ΔH/L
- Height dropped/runout length



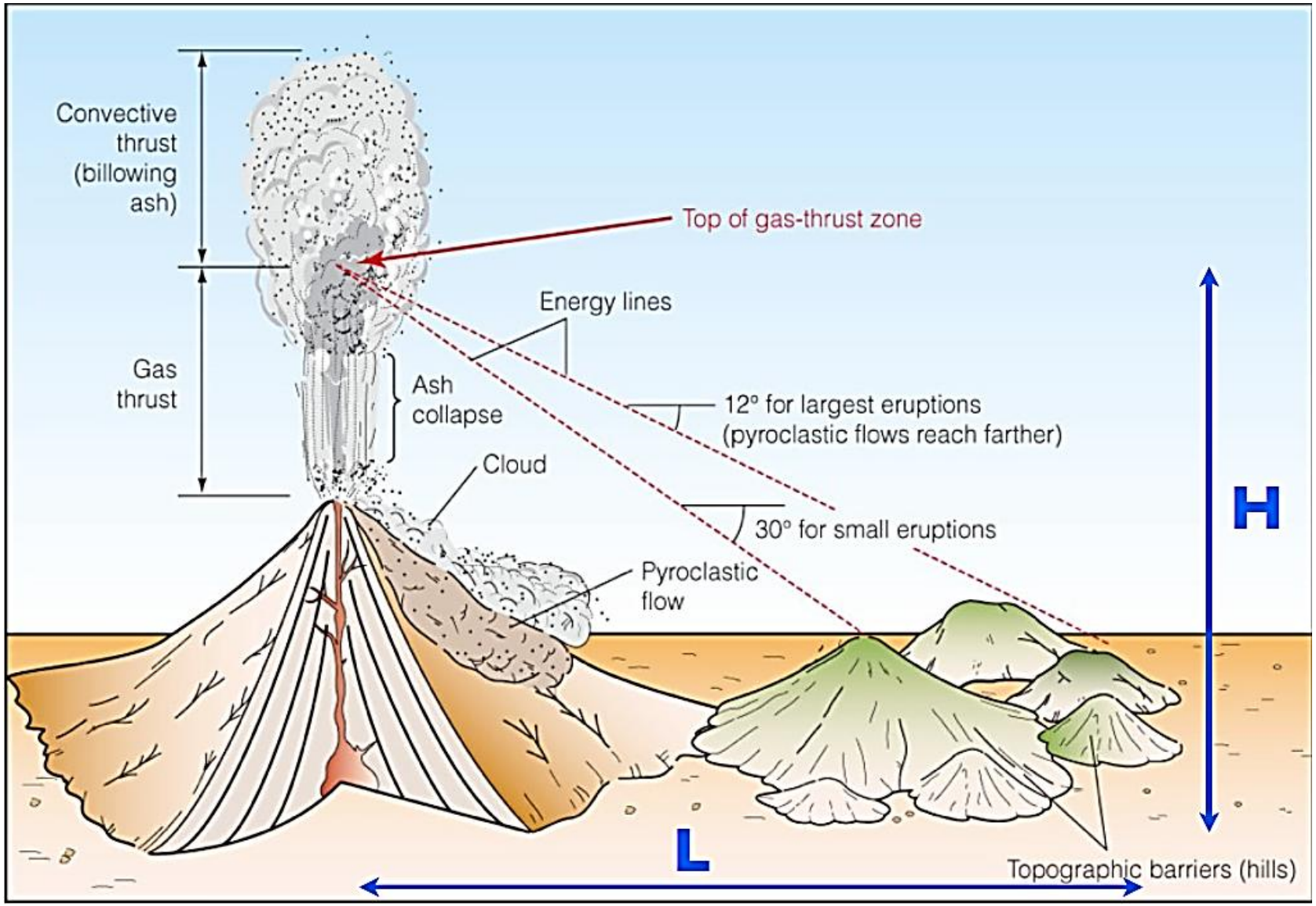
Block begins to slide when mgsinθ = μmgcosθ

Coeff. of friction, μ = $\frac{mgsin\theta}{mgcos\theta}$ = tanθ



ΔH/L or Heim coefficient

- For column-collapse flows, the H doesn't start at the top of the dome, it starts from where the collapse happens, at the top of the 'gas thrust' region
- Column-collapse PDCs usually collapse 500-2000 m above the summit



VDAP

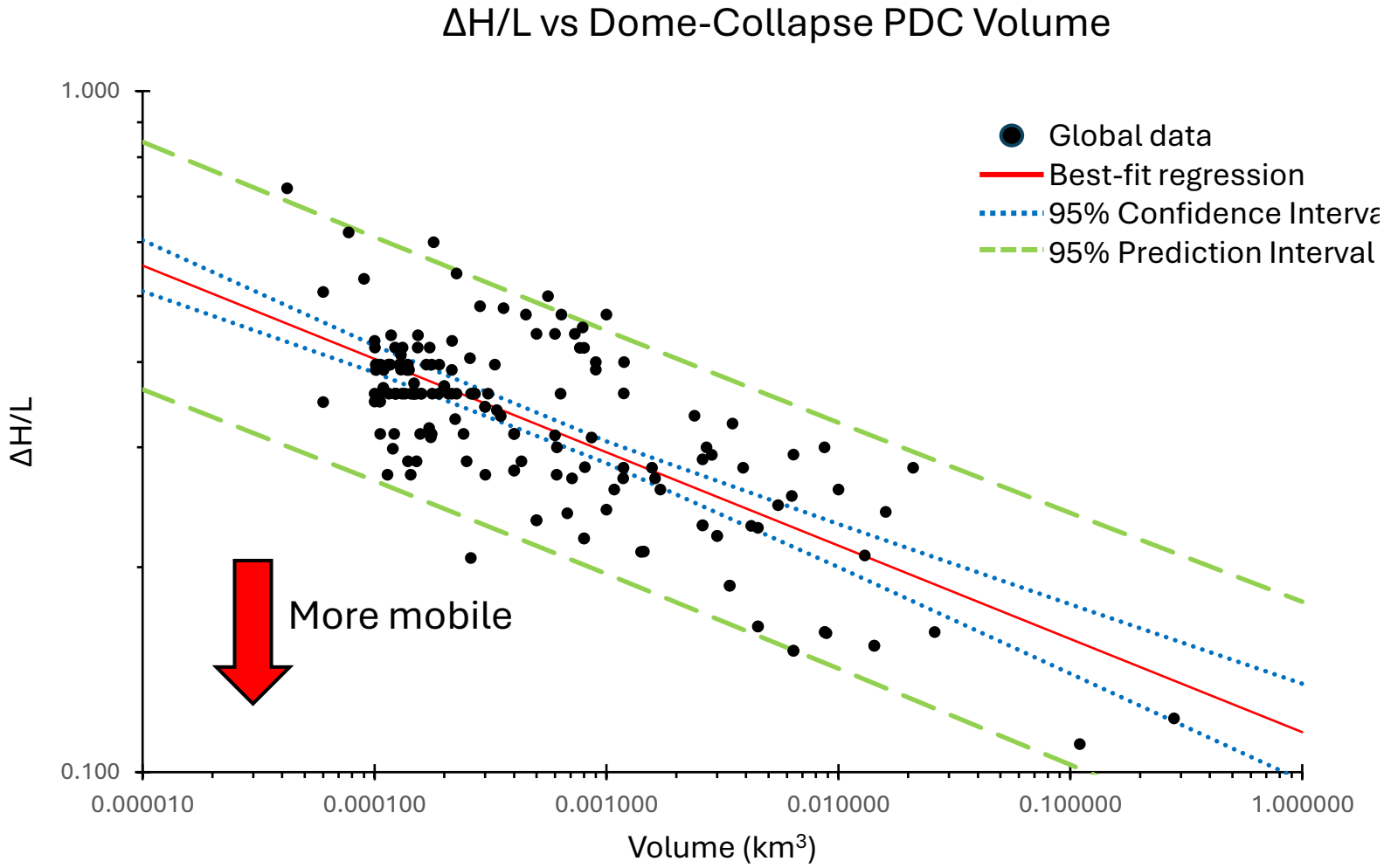
USGS

Why do we care about ΔH/L?

- ΔH/L can be used for the energy cone model for PDCs
- ΔH/L converted to degrees can be used for the basal friction angle in TITAN2D
- ΔH/L has a relationship with PDC volume, which means we can estimate what value to use in models for each volume we want to model

ΔH/L or Heim coefficient

- Should be independent of mass/volume, and depend only on material properties but actually ΔH/L shows an inverse relationship with volume
- Large volume flows are more mobile than small volume flows
- Lots of different theories to explain this 'excess mobility'. Large volume flows have complex rheologies that are difficult to simplify in any model.
- Lots of scatter in the data



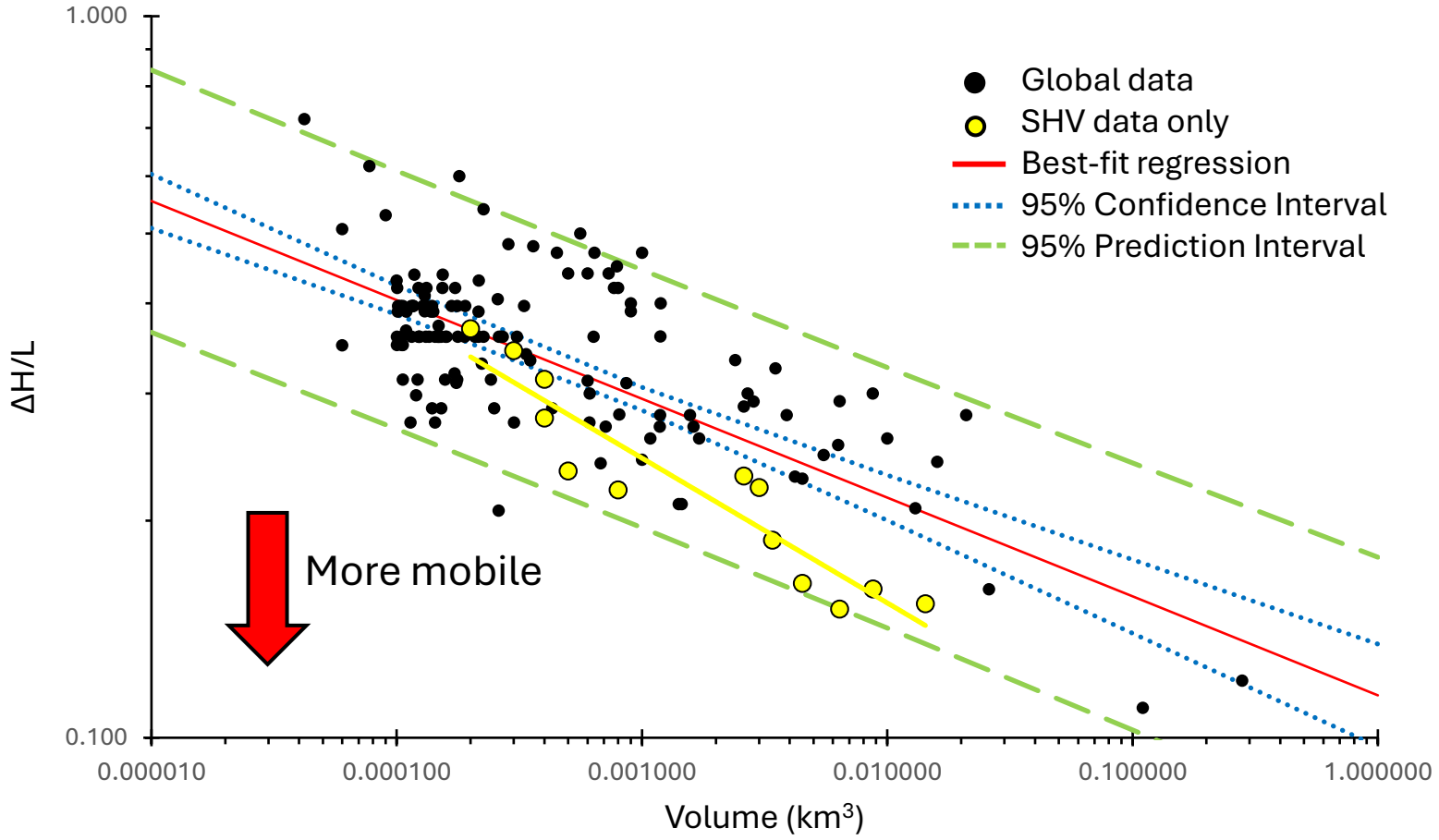
Best fit: $\Delta H/L = 0.120(\text{Volume}^{-0.127})$

VDAP

USGS

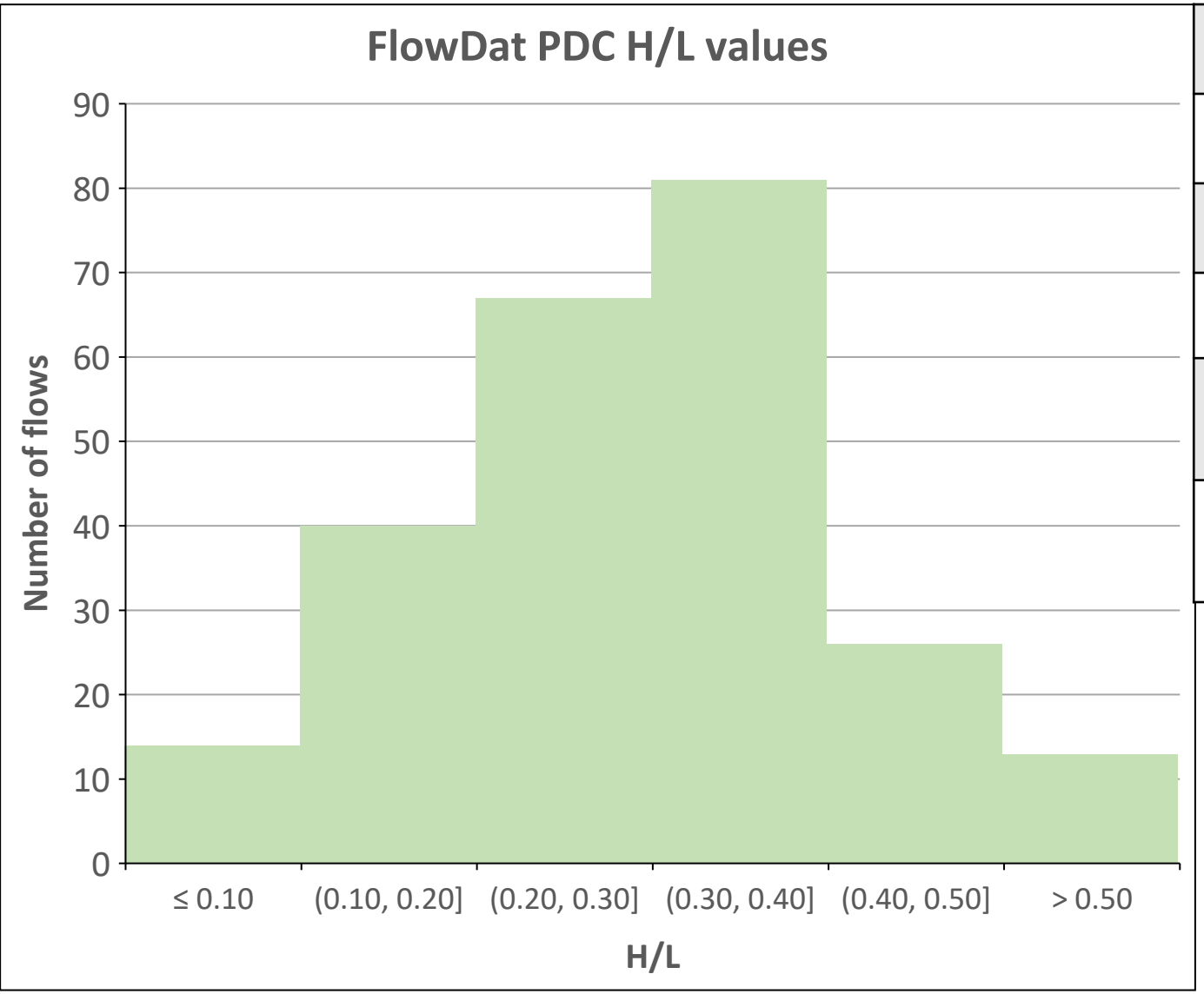
ΔH/L or Heim coefficient

- Different volcanoes can have differences in mobility
- For example, PDCs at SHV in Montserrat are more mobile than elsewhere, probably because of channelization and other topographic effects



VDAP

USGS



#	241 (with H/L values)
Avg.	0.30
Median	0.31
Min.	0.04 (very large ignimbrites)
Max.	0.72 (very small dome-collapses/rockfalls)
Middle 80%	0.15-0.42

Most PDCs: 0.15-0.4

VDAP

USGS

ΔH/L ranges for dome- and column-collapse PDC energy cones from FlowDat regressions

Volume (km^3)	Volume (10^6 x m^3)	ΔH/L best fit Dome collapse	ΔH/L best fit Dome collapse (degrees)	ΔH/L best fit Column collapse	ΔH/L best fit Column collapse (degrees)
0.0001	0.1	0.39	21.1	0.36	19.8
0.001	1	0.29	16.1	0.30	16.5
0.01	10	0.22	12.1	0.24	13.7
0.1	100	0.16	9.1	0.20	11.4

Dome-collapse PDCs (based on 121 flows): $\Delta H/L = 0.120\left(V^{-0.127}\right)$

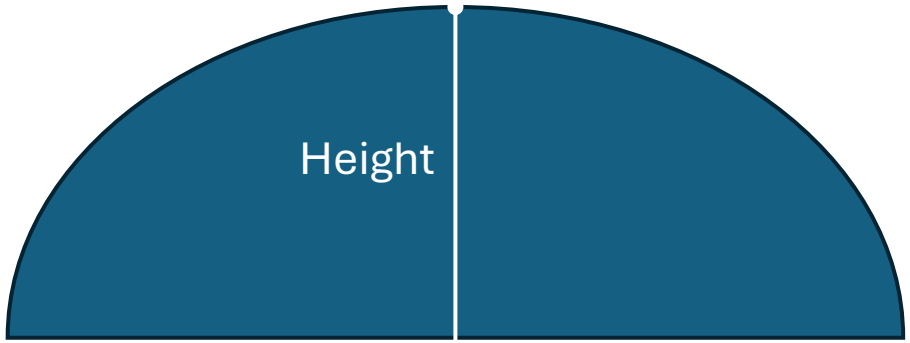
Column-collapse PDCs (based on 40 flows): $\Delta H/L = 0.166\left(V^{-0.084}\right)$

All PDCs (based on 188 flows): $\Delta H/L = 0.165\left(V^{-0.085}\right)$

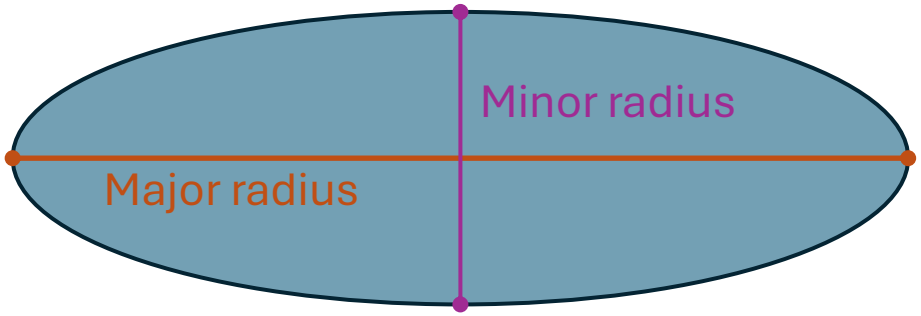
VDAP

USGS

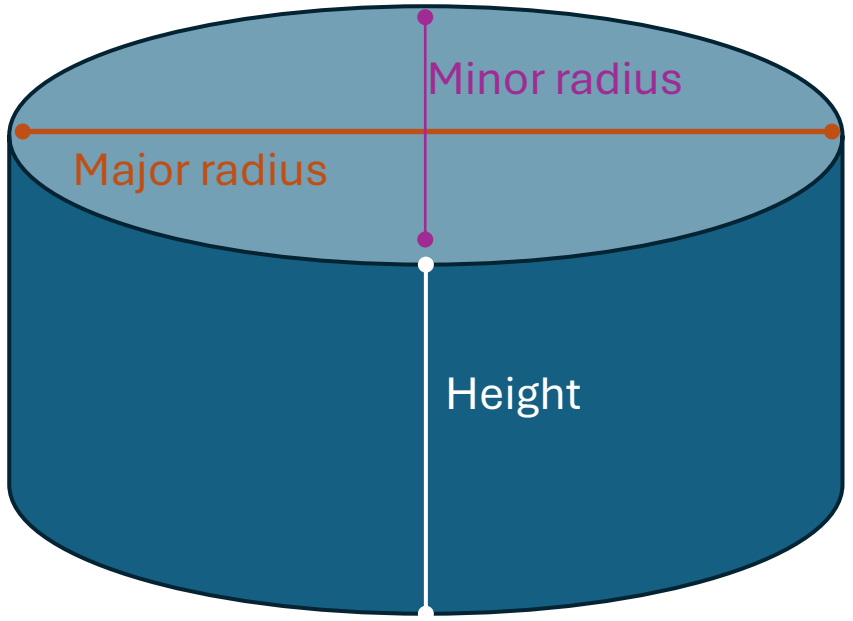
Source shape parameters: Pile size



Paraboloid (dome)



Paraboloid (dome)
Top view

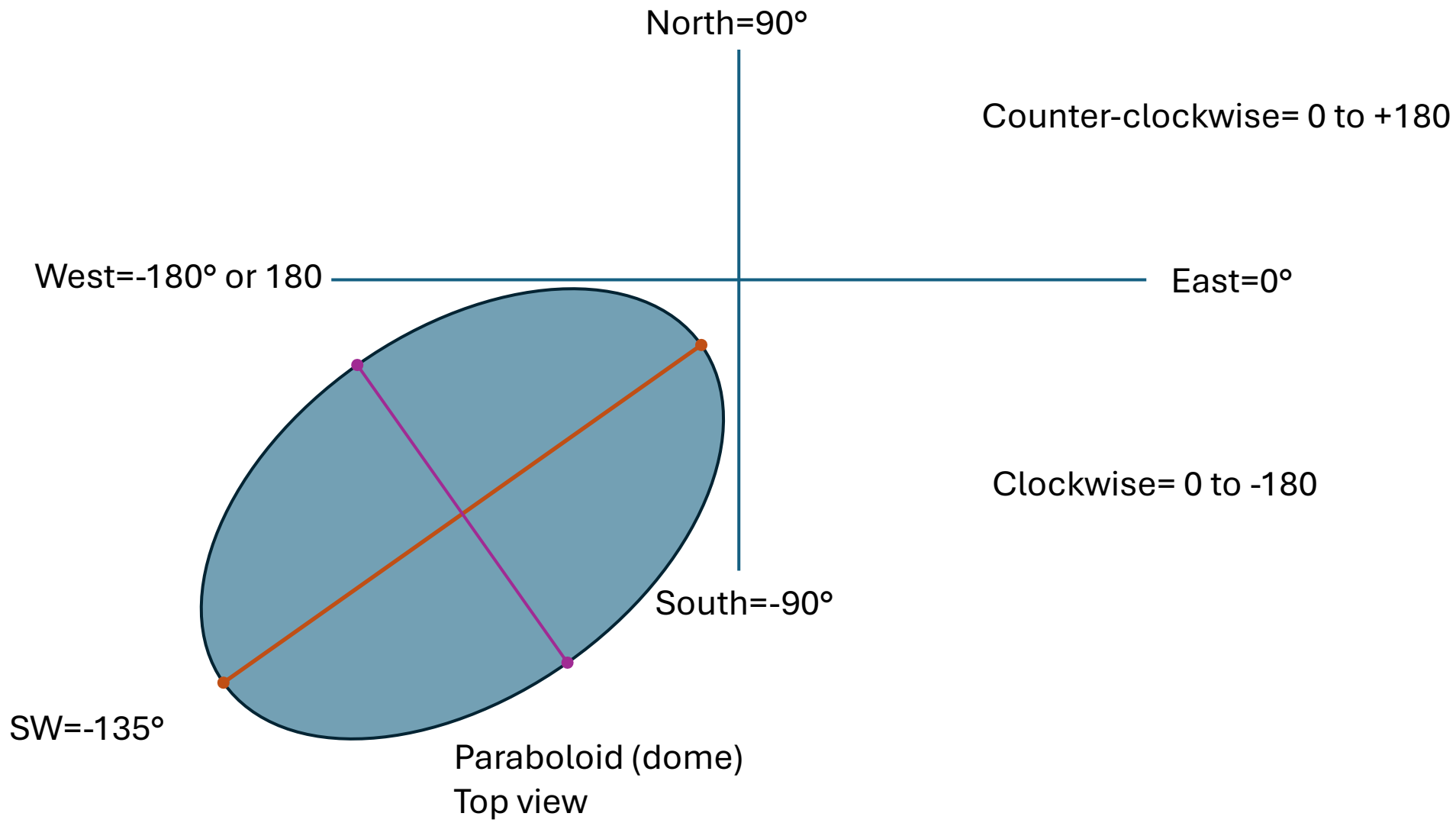


Cylinder (column)

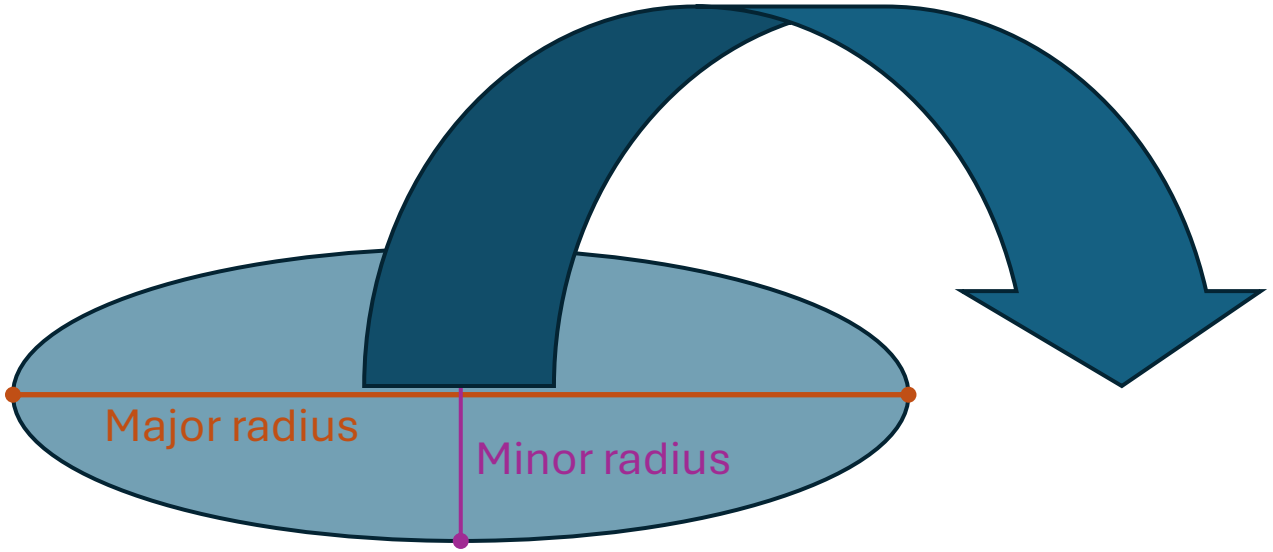
dome_volume_calculator.xlsx

Volume=1/2*pi*major_radius*minor_radius*height

Source shape parameters: Pile orientation



Source shape parameters: Flux source

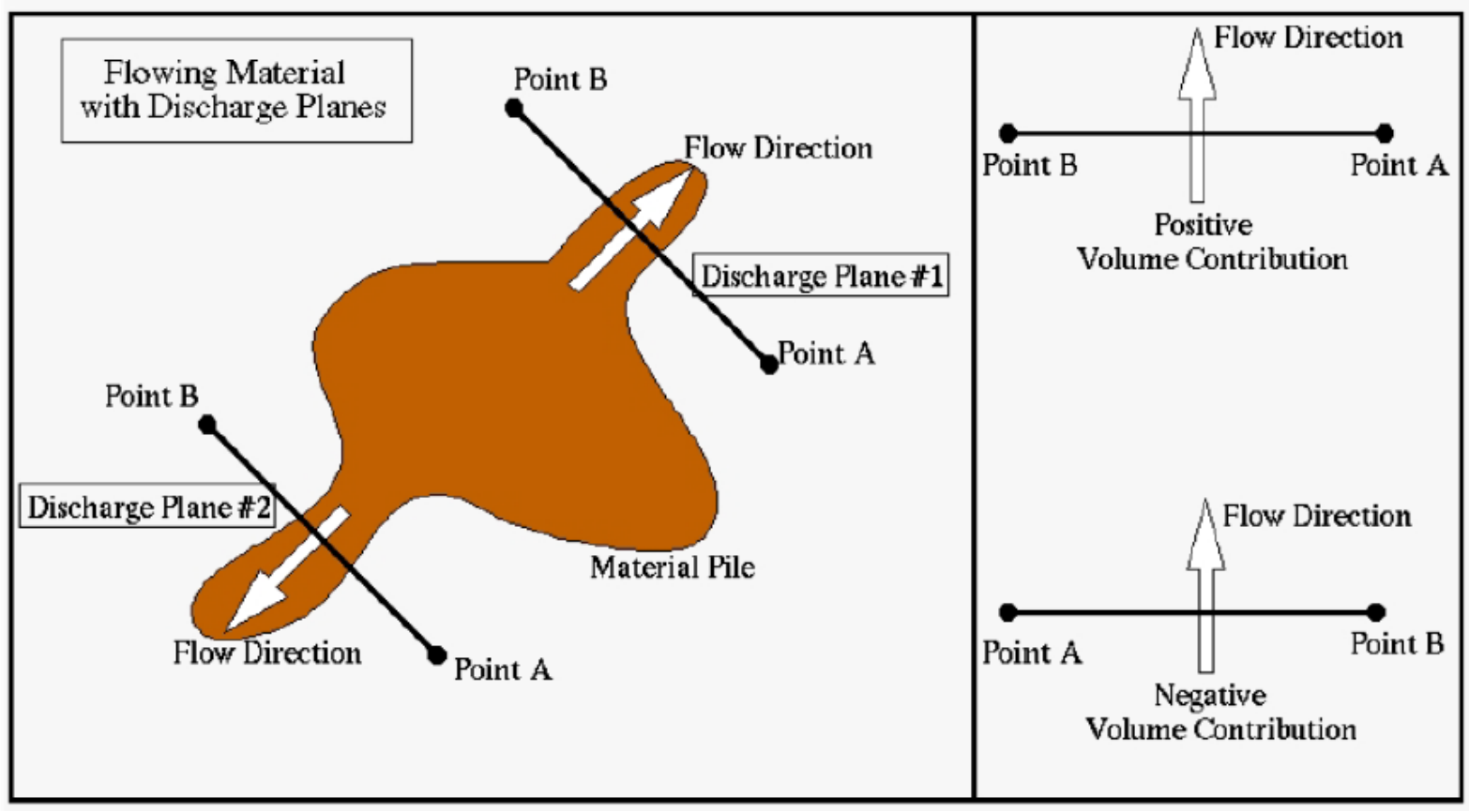


Paraboloid

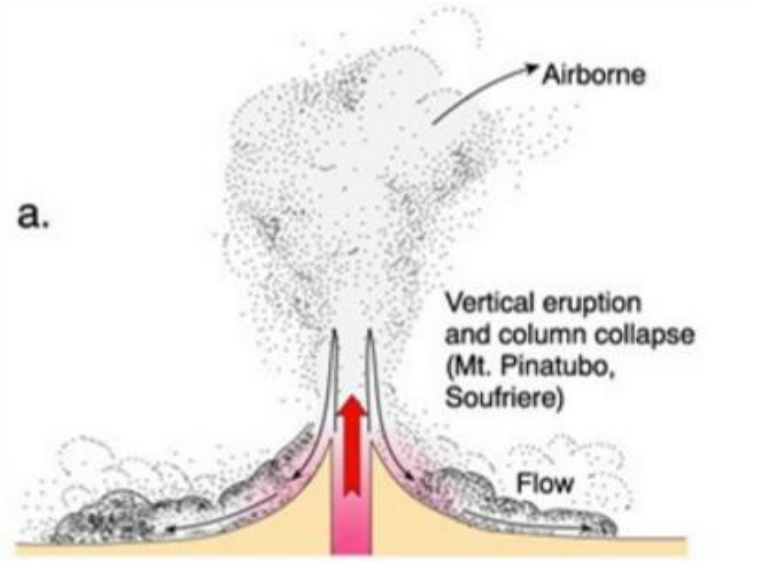
Still used a paraboloid shape, but instead of a pile height, it uses a flux (mass per unit area in meters per second) with a defined duration

Source shape parameters: Discharge Plane

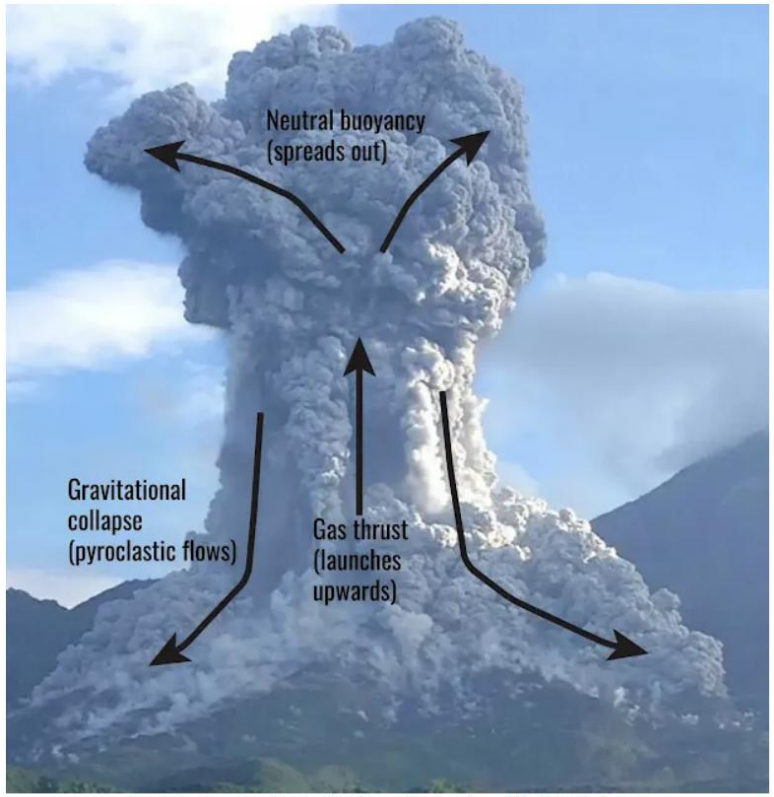
Allows you to set up “gates” to see how much volume passes through them



Direction



Winter (2001) after MacDonald (1972) and Fisher and Schminke (1984)



Pyroclastic Flows from Santiaguito Volcano Eruption, August 14, 2016, Guatemala.

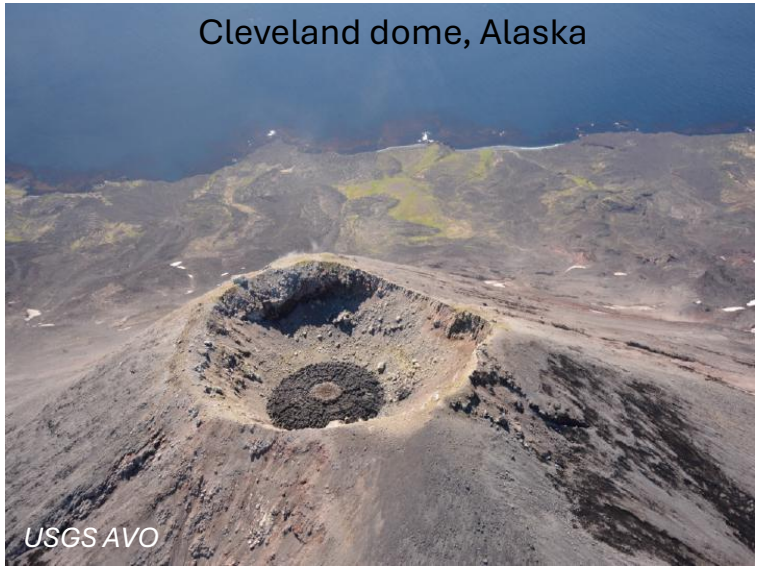
What contributes to dome-collapse PDC direction?

- Summit topography (horseshoe-shaped craters, breaches in craters, buttresses, etc.)
- Direction of growth of dome lobes/flows
- Height of dome
- Direction of previous collapses

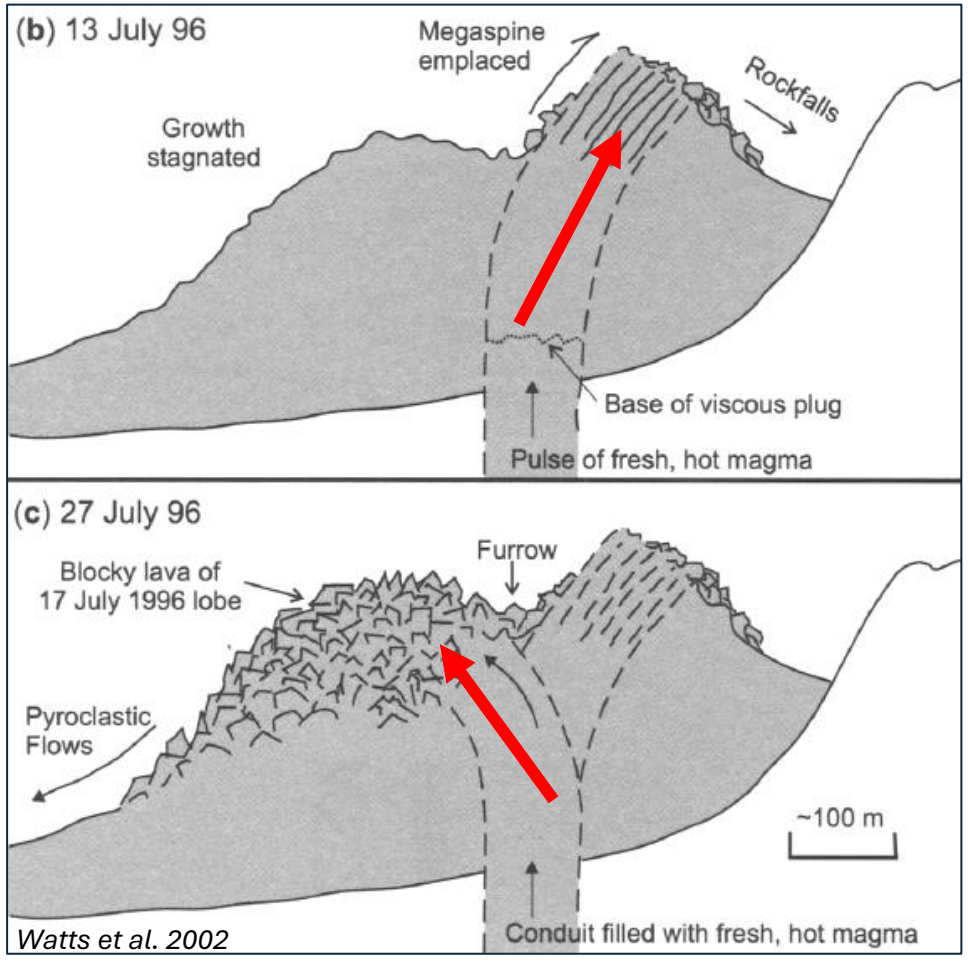
Breached crater, Merapi, Indonesia

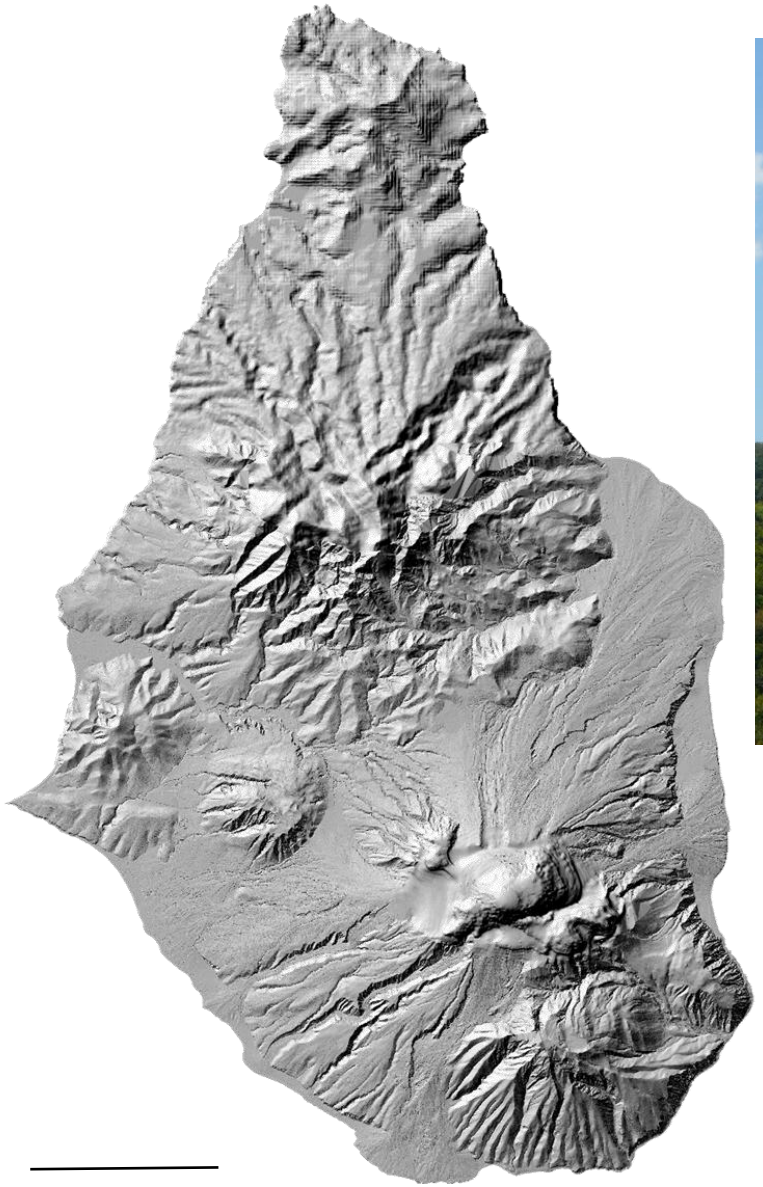


Cleveland dome, Alaska

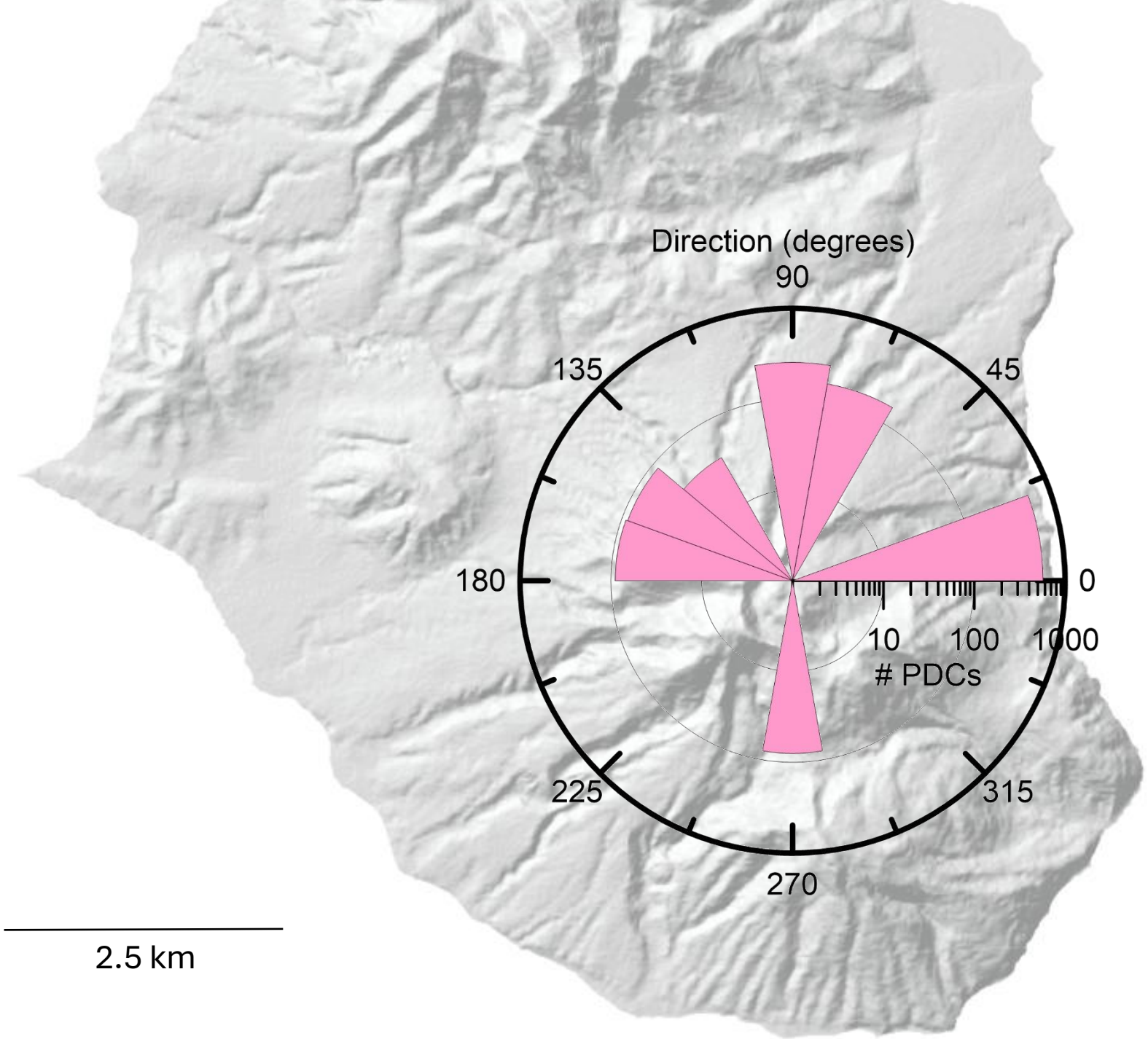


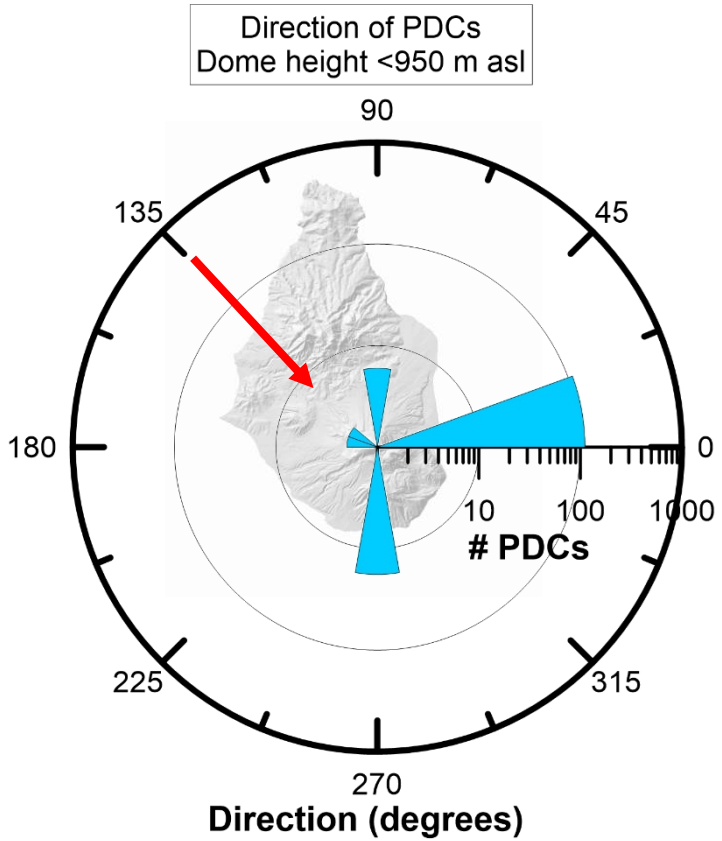
Growth of lobes at SHV, Montserrat

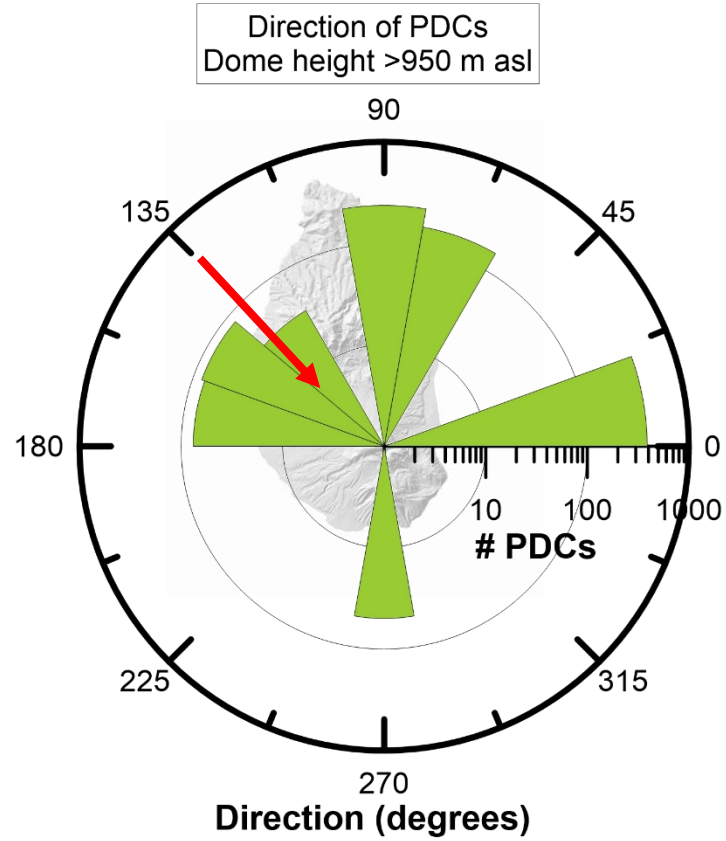


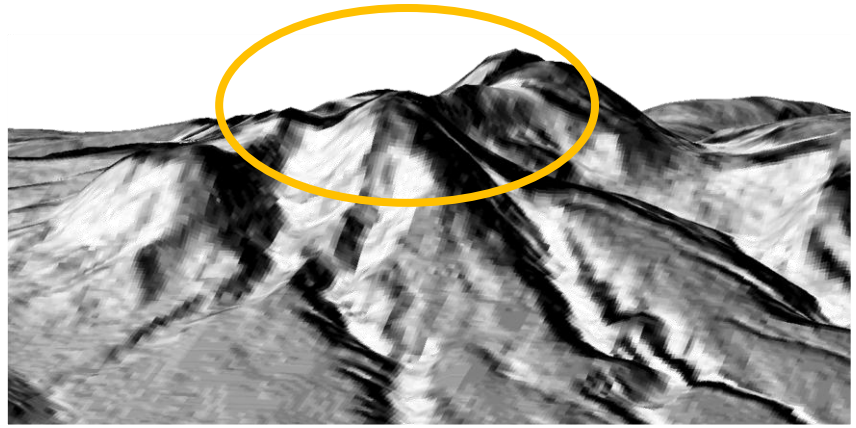

2.5 km

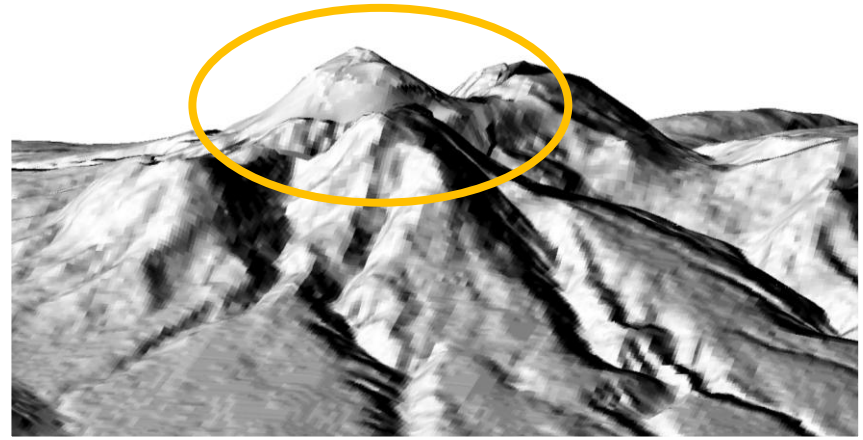


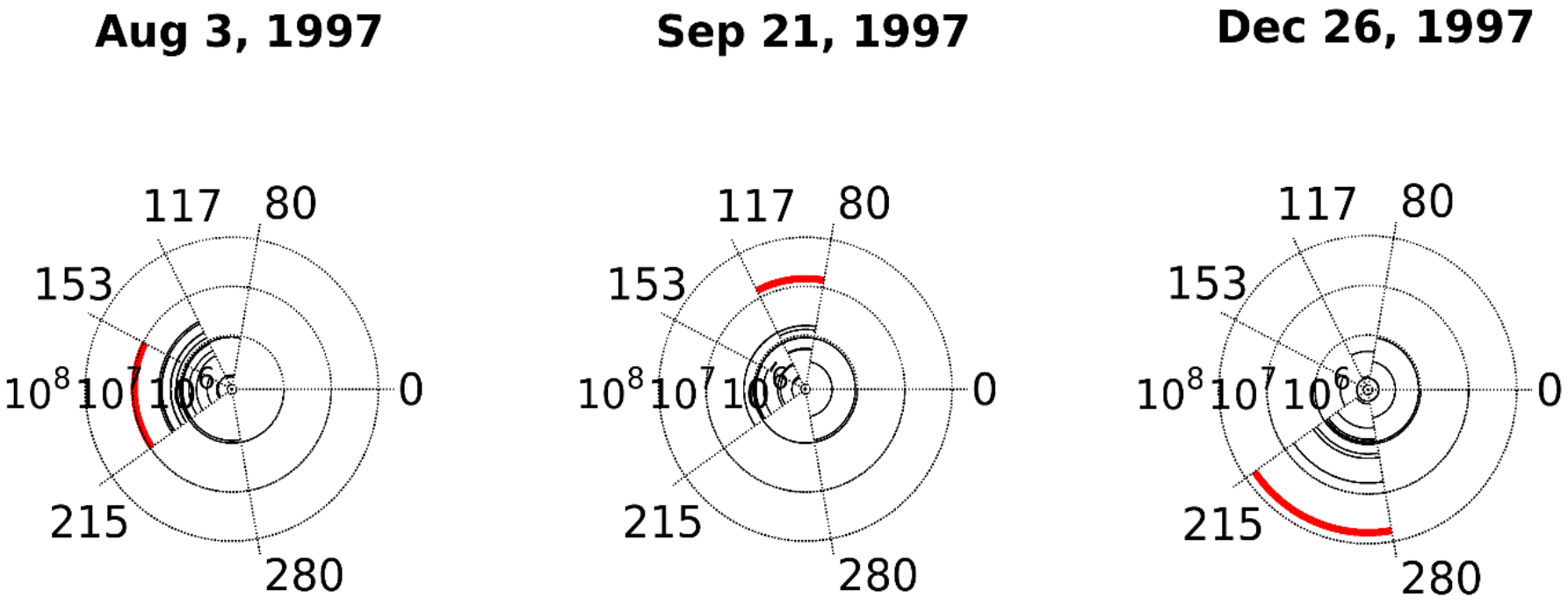

Direction (degrees)
90
135
45
180
0
225
315
270
10
100
1000
PDCs
2.5 km


Direction of PDCs
Dome height <950 m asl
90
135
45
180
0
225
315
270
10
100
1000
PDCs
Direction (degrees)


Direction of PDCs
Dome height >950 m asl
90
135
45
180
0
225
315
270
10
100
1000
PDCs
Direction (degrees)



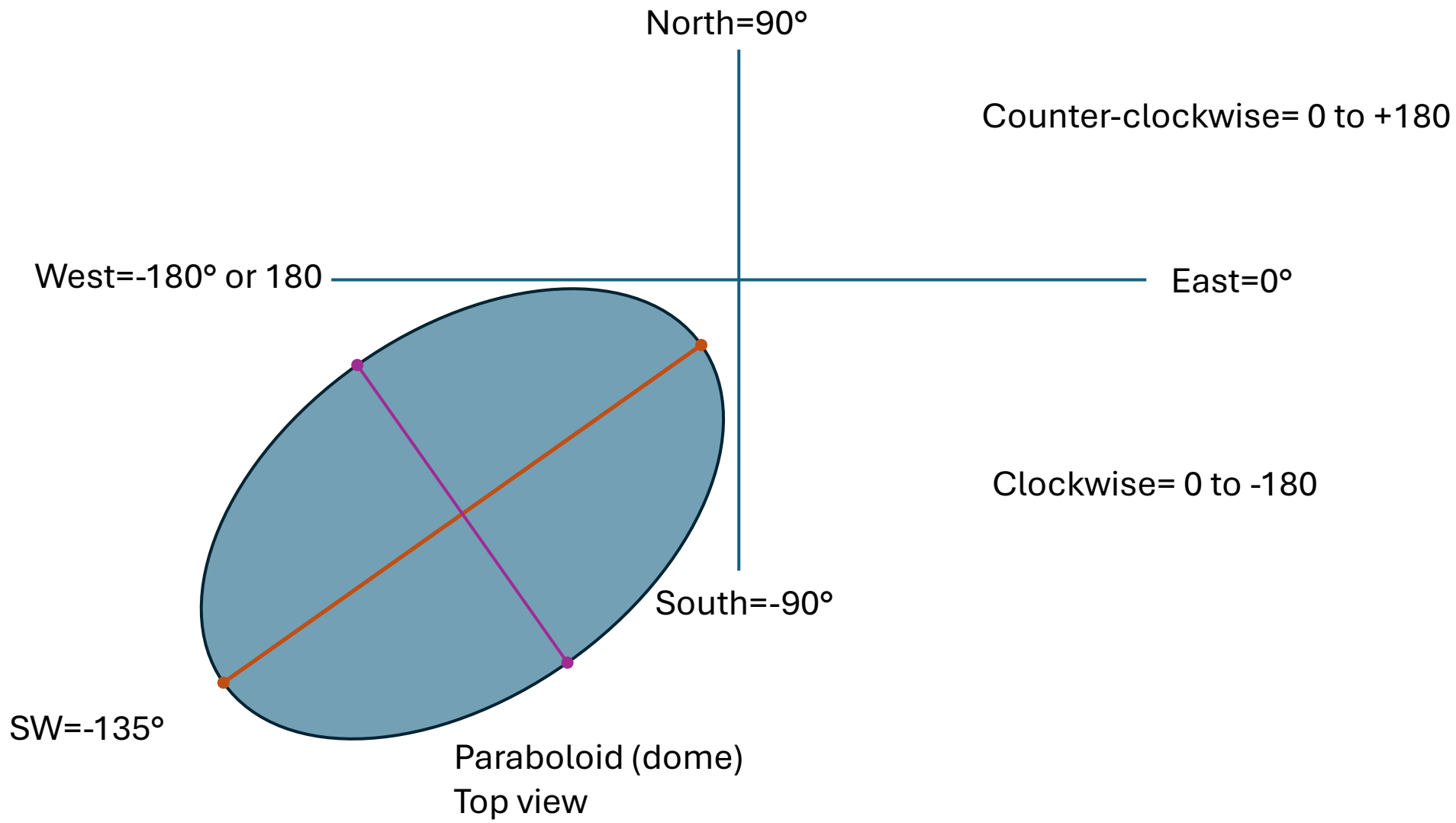




Wolpert et al. 2018

Series of smaller PDCs (black arc) often go in a ‘preferred direction’ for a period of time (probably related to direction of dome growth). After large PDCs (red arcs) occur, new PDCs often go in a different direction (Wolpert et al. 2018).

Source shape parameters: Pile orientation



Velocity

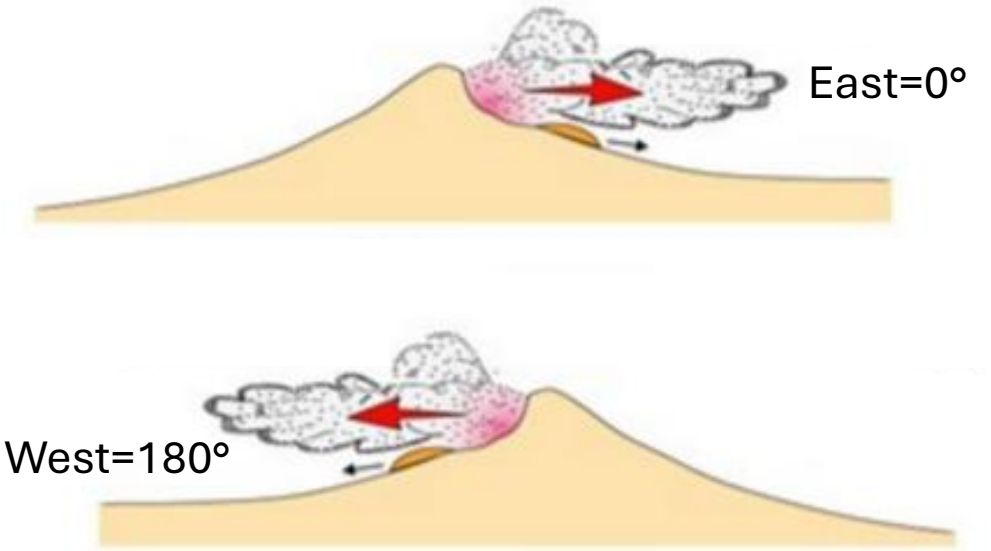


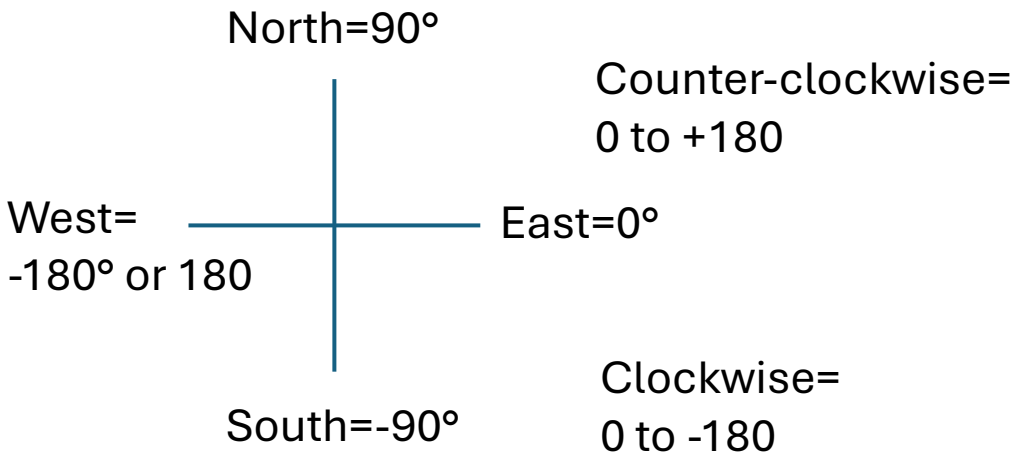
Mount St. Helens, USA



Winter (2001)

Initial direction (for velocity)





Dome-collapse direction

- Many times, it’s sufficient to use just the pile shape, placement, and orientation of the pile to get a dome to collapse in a specific direction (if desired, like in the case of a breached crater)
- Sometimes, if you know it should collapse in a specific direction but it’s not doing so, you can add a very small initial velocity and give that velocity a direction